

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

EE-6003-CBGS

B.E. VI Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Microprocessor and Microcontrollers

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Draw and explain architecture of 8086 microprocessor. 7
b) Write an assembly language program using 8086 instruction set to add following series upto seven terms:
 $1^2 + 3^2 + 5^2 + \dots$ 7
2. a) Explain flag registers of 8086 microprocessor. 7
b) Explain the purpose of stack in microprocessor. Explain stack structure exists in 8086 microprocessor. 7
3. a) Explain the architecture of programmable peripheral interface chip 8255. 7
b) Describe the DMA 8257 operation with diagram. 7
4. a) How will you interface 8255 with 8086. The address of port A is FOH. 7
b) Write an assembly language program to convert digital data to equivalent analog data using 8086 microprocessor. 7

EE-6003-CBGS

PTO

[2]

5. a) Describe the pin configuration of 8051 microcontroller. 7
b) List the registers of 8051 microcontroller and hence explain complete register set. 7
6. a) Explain the interrupts of 8051 microcontroller with priority status. 7
b) Describe the addressing modes of microcontroller 8051. 7
7. a) Explain the interfacing of stepper motor with 8051 microcontroller. 7
b) Design 8051 microcontroller based firing circuit. 7
8. Explain the following (any two): 14
 - a) USART 8251.
 - b) Instruction set of 8086.
 - c) Functional block diagram of 8096 microcontroller.
